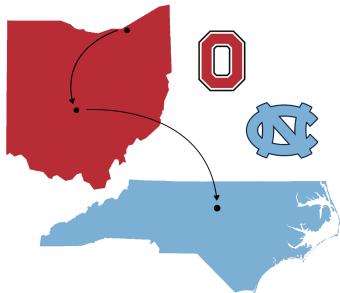


Assumptions, Information, and What We Learn from Different Models

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Research

Methods: causal inference, alternative randomized trial designs, estimating equations, data fusion

Applied: infectious diseases (HIV, respiratory), pharmacoepidemiology, social epidemiology



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Zivich PN *arXiv:2511.01960*

¹Footnotes are for asides or references

A Familiar Scenario(?)

Two research groups attempt to answer the same (a similar) question

- Use different study designs or methods
- End up with substantively different answers

Differences can arise even when research groups do everything 'right'

- Want to explain why we get different answers

Example: trial versus observational for hormone replacement therapy

- Trial found greater coronary heart disease, while observational found lower
- Differences due to underlying populations (e.g., time since menopause)²

²Hernán et al. *Epidemiology* 2008; 19(6):766-779

Two Traditions of Modeling

Statistical (phenomenological) modeling

- Use observations to estimate parameters of a model
- **Examples:** randomized trials, regression, inverse probability weighting

Mathematical (mechanistic) modeling

- Use a theoretical representation of a system through formulas that does not necessitate direct observations
- **Examples:** microsimulation, pharmacodynamic, agent-based models

Goal of This Presentation⁴

To better understand how we learn using quantitative methods

- Avoid talking past each other
- Better understand when and why we disagree

Provide a common language for thinking about statistical and mathematical models

- Borrow some conceptual devices from causal inference³
- Focus on assumptions to see what is learnable
- Examine an illustrative pharmacoepidemiology example

³Potential outcomes, Identification, Bounds

⁴Presentation based on Zivich *arXiv:2511.01960* 2025

Causal Inference

A_i : action for unit i , assume binary and occurs at baseline

Y_i : outcome, assume binary and occurs after baseline

W_i : other variables measured at or before baseline

$S_i = 1$: unit is member of population of interest

$O_i = (W_i, A_i, Y_i, S_i = 1)$: observed data from member of population of interest

$I(\cdot)$: indicator function, return 1 if argument is true and 0 otherwise

$\Pr_{S=1}(\cdot)$: probability function for a variable in the population of interest.⁵

Subscript i may be left off for convenience

⁵I am using $\Pr_{S=1}(\cdot)$ as shorthand for $\Pr(\cdot \mid S = 1)$

Defining Causal Effects

Rely on *potential outcomes*

$$Y_i^a$$

which denotes the value of the outcome Y if unit i were to take action a

Can think of each person as having two potential outcomes

- Y_i^1 : outcome if i took action 1
- Y_i^0 : outcome if i took action 0

Interest Parameter

Causal Risk Difference

Risk if everyone took action 1

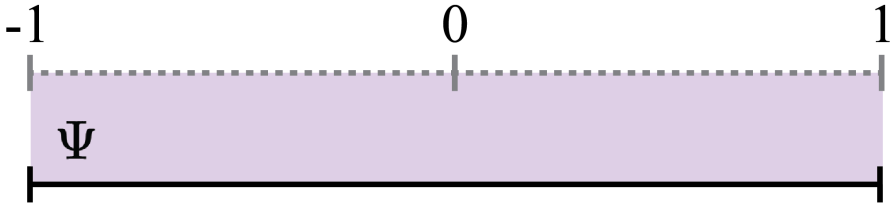
$$\psi := \Pr_{S=1}(Y^1 = 1) - \Pr_{S=1}(Y^0 = 1)$$

Risk if everyone took action 0

Parameter Space, Ψ

- All the values that ψ could be
- Causal risk difference must be in $[-1, 1]$

Parameter Space



All methods seek to shrink the possible interval for the causal effect from $[-1, 1]$

Can only learn more by making assumptions⁶

Identification: expression of causal parameter as a function of available information

- Clarifies the additional assumptions being made

⁶ “[T]he questions that we raise and our doubts depend on the fact that some propositions are exempt from doubt, are as it were like hinges on which those turn. [...] If I want the door to turn, the hinges must stay put” - Ludwig Wittgenstein *On Certainty* §341-343

A Short Review of Identification with Statistical Modeling

Statistical modeling would begin with collection some observations

- Data $O_i = (W_i, A_i, Y_i)$ from n observations

What can we learn from these observations?

- Depends on our assumptions

Some Starting Identification Assumptions

Assume

1. **Temporality**: variables occur in the following time-order $W \rightarrow A \rightarrow Y$
2. **Independent and Identically Distributed**: random sample of n units from our target population $S = 1$
3. **No Measurement Error**: all variables in O_i are correctly measured for all n units
4. **Causal Consistency**: if $a := A_i$ then $Y_i^a = Y_i$, which provides a bridge or connection between the *potential* outcomes and *observed* outcomes⁷

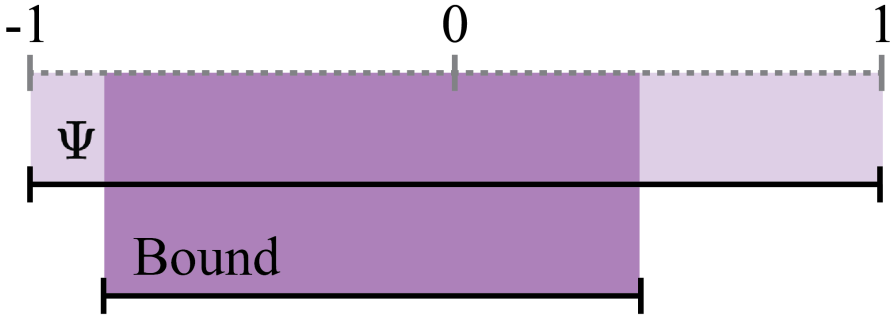
What can we learn under these relatively basic assumptions?

⁷Cole & Frangakis *Epidemiology* 2009; 20(1):3-5

It turns out that we can learn quite a bit

- Allow us to bound the possible causal effect
- From $\Psi = [-1, 1]$ to a range with a width of 1
 - *Halve* the set of possible values
- No information needed about study design
 - No discussion about confounding, parametric models, etc.

Nonparametric Bounds



An Aside About Confidence Intervals

Bounds are distinct from Confidence Intervals (CIs).⁸

- Bounds capture *epistemic* uncertainty
 - Systematic, lack of knowledge about something
 - Uncertainty from assumptions
- CIs capture *random/statistical* uncertainty
 - Fluctuations due to random sampling
 - Uncertainty from noise

Intuition

- Increasing sample size shrinks CIs but won't change the bound width

⁸Vansteelandt et al. *Statistica Sinica* 2006; 16(3):953–979

Moving Beyond the Bounds

Limitations

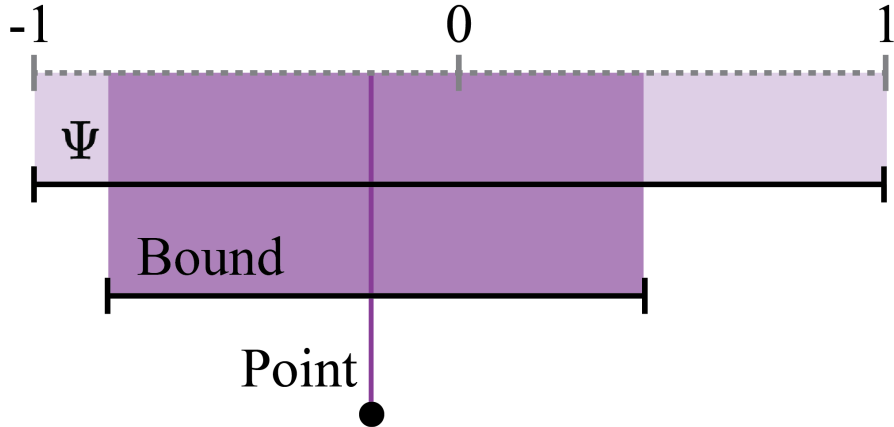
- Still unsatisfyingly wide, always contain $\psi = 0$, no causal effect

To narrow the bounds, need additional assumptions

- Marginal exchangeability of A and Y^a
 - Randomized trial
- Conditional exchangeability of A and Y^a given W
 - Observational, no unmeasured confounding
- An instrumental variable Z
- Others (e.g., proxy variables for an unmeasured confounder)

These auxiliary assumptions allow us to identify a unique *point*

Partial and Point Identification



Example of Point Identification

If we also assume (or know) that

- 5. **Study Design:** data O_i came from an observational study
- 6. **No unmeasured confounding:** W_i is a sufficient adjustment set⁹

Then the causal effect can be point identified via

$$\Pr_{S=1}(Y^a = 1) = \sum_{w \in \mathcal{W}} \Pr_{S=1}(Y = 1 \mid A = a, W = w) \Pr_{S=1}(W = w)$$

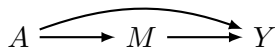
⁹More formally, $Y^a \perp\!\!\!\perp A \mid W$ and $0 < \Pr(A = a \mid W = w) < 1$ for all $w \in \mathcal{W}$

Extending Identification to Mathematical Models

Mathematical modeling would begin with specifying a mechanism

- Base this mechanism on ‘theory’
- No directly observed data is required

For exposition, consider simple mechanistic model¹⁰



- For ease, suppose all binary variables

¹⁰Note that this structure implies **temporality** and **no measurement error**

Remains the causal risk difference, ψ , but different decomposition based on mechanism

$$\Pr_{S=1}(Y^a = 1) = \sum_{m \in \{0,1\}} \Pr_{S=1}(Y^a = 1 \mid M^a = m) \Pr_{S=1}(M^a = m)$$

Diagram illustrating the decomposition of the causal risk difference into two components:

- Parameter:** $\Pr_{S=1}(Y^a = 1)$ (Leftmost term, highlighted in light blue)
- Probability of outcome Y given A and M :** $\Pr_{S=1}(Y^a = 1 \mid M^a = m)$ (Middle term, highlighted in light green)
- Probability of intermediate M given A :** $\Pr_{S=1}(M^a = m)$ (Rightmost term, highlighted in light purple)

These represent the *true* probability functions

Using a Mathematical Model

We specify some parameterized functions for the mechanism

The diagram illustrates the mathematical model for ψ . It features the equation $\bar{\psi}(\theta, \lambda) := \sum_{m \in \{0,1\}} h(a, m; \lambda) g(a; \theta)$. Three colored boxes highlight the components: a blue box for $\bar{\psi}(\theta, \lambda)$, a green box for $h(a, m; \lambda)$, and a purple box for $g(a; \theta)$. Annotations with arrows explain each part: a blue arrow points from 'Math model for ψ ' to the blue box; a green arrow points from 'Function for outcome Y given A and M ' to the green box; and a purple arrow points from 'Function for intermediate M given A ' to the purple box.

$$\bar{\psi}(\theta, \lambda) := \sum_{m \in \{0,1\}} h(a, m; \lambda) g(a; \theta)$$

Math model for ψ

Function for outcome Y given A and M

Function for intermediate M given A

Identification: assumptions that relate the chosen functions and their parameters, (θ, λ) , to the true probabilities

The assumed structure already has **temporality** and **measurement error** built in

Some additional restrictions on h and g

1. Both functions return values in $[0, 1]$ for parameter spaces Θ, Λ .¹¹
2. Output of g (h) covers $[0, 1]$ for all unique values of a (and m) for Θ (Λ)¹²

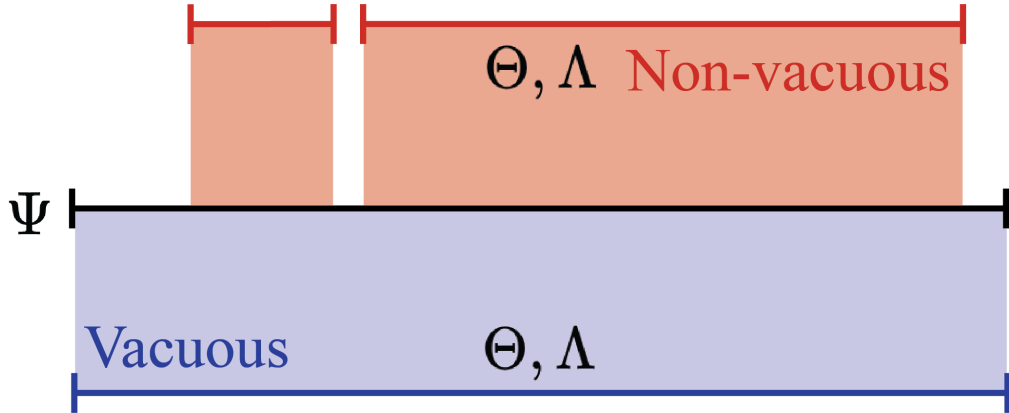
Why: ensures that $\bar{\psi}(\theta, \lambda)$ covers all of Ψ for valid parameters

- Ruling out possible ψ values is done through choices of θ, λ , not g, h
- Call math models that meet this property *vacuous* models

¹¹Formally, $g : \Theta \rightarrow [0, 1]$, $h : \Lambda \rightarrow [0, 1]$

¹²Formally, $\{g(a; \theta) : \theta \in \Theta\} = [0, 1]$, $\{h(a, m; \lambda) : \lambda \in \Lambda\} = [0, 1]$ for all unique values of a, m

Vacuous vs Non-vacuous Models



Now we simply have to select values of θ, λ

Given h, g , suppose we could ask an oracle¹³ to reveal the correct (true) parameter choices for these functions, denoted as θ^*, λ^* . Then for all unique a, m

$$g(a; \theta^*) = \Pr_{S=1}(M^a = 1)$$

$$h(a, m; \lambda^*) = \Pr_{S=1}(Y^a = 1 \mid M^a = m)$$

Then it follows that $\psi = \bar{\psi}(\theta^*, \lambda^*)$

- A *vacuous* model ensures θ^*, λ^* exist

¹³or genie, or sphinx, or your favorite other mythical being with magic insight

Knowledge of θ^*, λ^* is difficult.¹⁴ Consider partial identification instead

- Subset of possible values for θ^* , denoted by Θ^*
- Subset of possible values for λ^* , denoted by Λ^*

Model Capture: for all unique a, m assume that

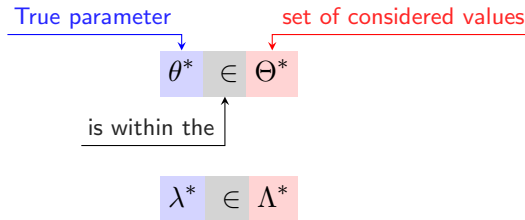
The diagram illustrates the concept of partial identification. It features two horizontal bars, each divided into three segments: a light blue segment on the left, a light gray segment in the middle, and a light red segment on the right.

The top bar represents the true probability. A blue arrow labeled "True probability" points to the blue segment, which contains the expression $\Pr_{S=1}(M^a = 1)$. A red arrow labeled "possible outputs for function" points to the red segment, which contains the set $\{g(a; \theta) : \theta \in \Theta^*\}$. A black arrow labeled "is within the" points from the blue segment to the gray segment, which contains the symbol $\in$.

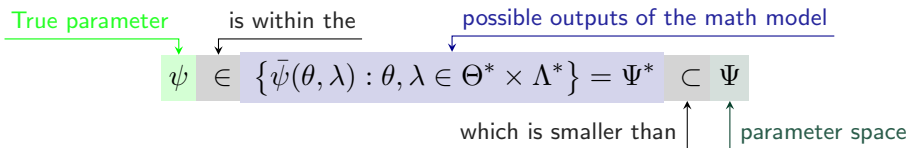
The bottom bar represents the conditional probability. The blue segment contains the expression $\Pr_{S=1}(Y^a = 1 \mid M^a = m)$, and the red segment contains the set $\{h(a, m; \lambda) : \lambda \in \Lambda^*\}$. The gray segment contains the symbol $\in$.

¹⁴If not unreasonable outside the world of examples we can create

Another (maybe easier) way to specify the **Model Capture** assumption

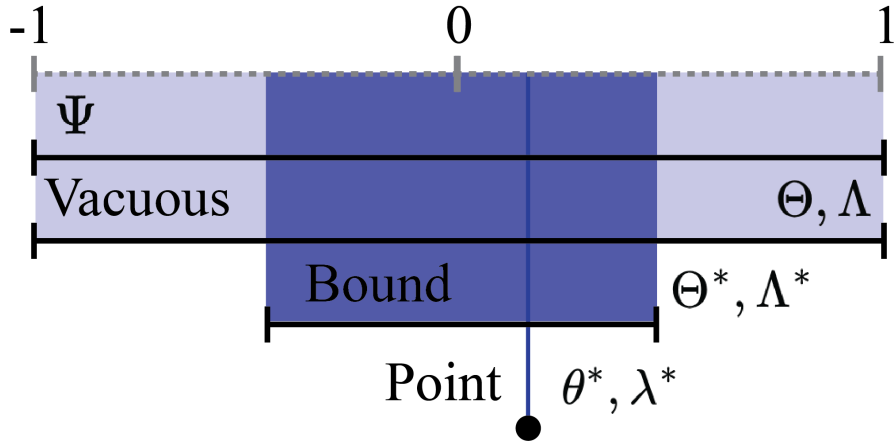


Model Capture allows for



In other words, we can compute bounds for ψ

Identification Visualization



Vacuous model

- Ought to prefer vacuous models when we can construct them
- Analogous to parametric vs nonparametric identification in statistical modeling¹⁵

Width of mathematical model bounds

- Function of choices for Θ^*, Λ^*
- Unlike statistical modeling, no guaranteed size reduction in bounds
- Like statistical modeling, information needs to be relevant for $S = 1$

¹⁵The g-null paradox for the g-formula is one such example

Motivating Example

Question: effect of amlodipine (10 mg)¹⁶ vs placebo on hypertension status at 24 hours?

- Population: adults with a SBP ≥ 140 in the US

Notation

- A : treatment (1: amlodipine, 0: placebo)
- Y : hypertension status¹⁷ 24 hours after baseline
- $S = 1$: US adults with a SBP ≥ 140
- ψ : corresponding causal risk difference

¹⁶ A calcium channel blocker used to treat high blood pressure and chest pain

¹⁷ Defined as a systolic blood pressure of ≥ 140 mm Hg

Can learn ψ by

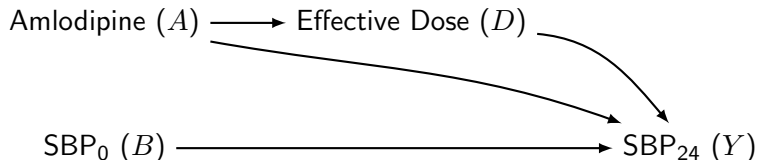
- Conducting a randomized trial
- Use existing observational data and rely on some unverifiable assumptions (e.g., no unmeasured confounding)
- Rely on only the bounds

Regardless of approach taken, identification allows us to more clearly articulate assumptions

No direct observations on (A, Y) and instead rely on theory

- External information or beliefs about amlodipine and SBP

Assume structure between different variables



Pharmacodynamic model

$$\bar{\psi}(\gamma, \theta, \lambda) = \int_{\mathcal{B}} h(b, a, g(a, \theta); \lambda) f(b; \gamma)$$

Probability density function for blood pressure, γ

- Determined by the population

Function for effective dose, θ

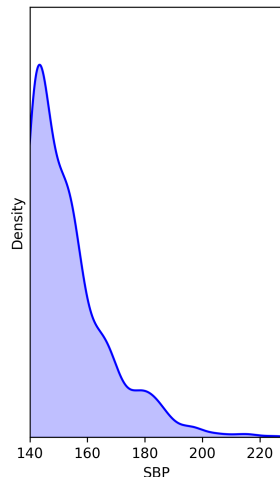
- Active drug bioavailable at 24 hours given a

Function for hypertension status, λ

- Return hypertension given b, a, d

$f(b; \gamma)$ – Probability density function for blood pressure

- Leave as generic probability density function
- **External Information:** Need SBP for US adults with SBP > 140
 - γ based on empirical distribution of the first measured SBP from NHANES¹⁸ 2017-2018



¹⁸National Health and Nutrition Survey

$g(a; \theta)$ – effective dose

- Simple proportion model: $g(a, \theta) = \theta_0 \times \theta_1$
 - θ_0 : dose of amlodipine, θ_1 : percent of available drug at 24 hours
 - Non-negative values will always return non-negative effective dose
- **External Information:**
 - $\theta_0 = 10$ when $A = 1$ and zero otherwise
 - $\Theta_1^* = [25\%, 40\%]$ based on dosing study¹⁹

¹⁹Kim et al. *Clinical Therapeutics* 2013; 35(7):934-940

$h(b, a, m; \lambda)$ – hypertension status

- $h(b, a, m; \lambda) = I(\bar{y}(b, a, m; \lambda) < 140)$, with E-max model

$$\bar{y}(b, a, m; \lambda) = b - \left[\lambda_0 + a\lambda_1 \left(\frac{m}{\lambda_2 + m} \right) + a\lambda_3 \right]$$

External Information:

- λ_0 : constant change in B , $\lambda_0^* := 0$
- λ_1 : maximum response, $\Lambda_1^* := [16.3, 36.3]$.²⁰
- λ_2 : ED50, $\Lambda_2^* := [0.1, 13.0]$.²¹
- λ_3 : effect of amlodipine not through effective concentration, $\lambda_3^* := 0$

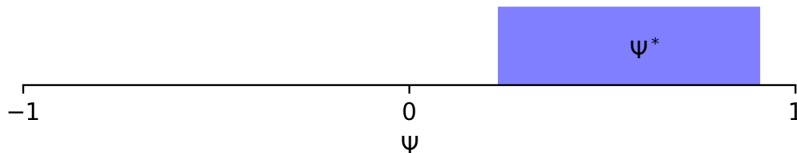
²⁰Park et al. *Basic Clin Pharmacol Toxicol*, 2019; 125(4):345-352

²¹Heo et al. *British Journal of Clinical Pharmacology* 2016; 82(6):1557-1567

Results

Applying the model, we get $\Psi^* = [0.23, 0.91]$

- Amlodipine is effective at reducing hypertension status



But can we trust these results

- Review the identification assumptions being made

Evaluation of the Mathematical Model

Model Capture:

- $f(b; \gamma)$
 - Assumes NHANES 2017-2018 *exactly* represents SBP distribution for $S = 1$
- $g(a; \theta)$
 - No manufacturing error with dosage
 - Available drug info comes from male South Koreans with SBP < 150 .²²
- $h(b, a, m; \lambda)$
 - $\lambda_0 := 0$ means no natural variation in hypertension over 24 hours
 - Choices for λ_1, λ_2 also comes from South Koreans with SBP < 150
 - $\lambda_3 := 0$ means side-effects (e.g., nausea, fatigue) do not alter behaviors

Summary: concerns about information sources, bounds likely too narrow

²²Body weight distributions are also substantially different (mean: 68 kg South Korean, 84 kg NHANES)

Conclusions

Identification provides framework to compare statistical and mathematical modeling

- Focus on bounding causal effects
- Help formalize assumptions and how to talk about them
- Opens up analogies between the two traditions

Illustrated with a simple pharmacodynamic model

- Highlights issues for the model
- Concrete suggestions on how to improve the model

Provides a way to communicate across fields

- Helpful to be comfortable with both traditions for infectious disease epidemiology
- More holistic view of how we can address pressing scientific questions

Framework allows for traditions to be jointly used to address a question

- My own work uses mathematical models for extrapolation under uncertainty with statistical models, and to address non-positivity²³
 - Framework more clearly lays out assumptions being made
- Can re-examine other hybrid models

²³Zivich et al. *Epidemiology* 2024; 35(1):23-31, Zivich et al. *JECH* 2026; In-Press

Extend from bounds to distributions for ψ

- Reflect amount of (un)certainty in information on θ, λ

Extend to allow for interactions between units

- Major use of mathematical models
- Can be done under an interference framework from causal inference²⁴

Actual application to build or compare mathematical models

²⁴Hudgens & Halloran *JASA* 2008; 103(482):832-842

Questions?



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Zivich PN *arXiv:2511.01960*

Appendix

Constructing the Bounds

It follows that

$$\begin{aligned}\Pr_{S=1}(Y^1 = 1) &= \Pr_{S=1}(Y^1 = 1 \mid A = 1) \Pr_{S=1}(A = 1) + \Pr_{S=1}(Y^1 = 1 \mid A = 0) \Pr_{S=1}(A = 0) \\ &= \Pr_{S=1}(Y = 1 \mid A = 1) \Pr_{S=1}(A = 1) + \Pr_{S=1}(Y^1 = 1 \mid A = 0) \Pr_{S=1}(A = 0)\end{aligned}$$

Law of total probability

Causal consistency

All quantities in terms of O_i , except $\Pr_{S=1}(Y^1 = 1 \mid A = 0)$

- But we know it must be within $[0, 1]$

Synthesis: Case Study in Transportability

Trial on impact of 2- vs 1-drug therapy on 20-week CD4 (marker of disease progression) in women with HIV²⁵

- Average causal effect comparing 2- vs 1-drug on 20-week CD4 in target population
- ACTG 175 (trial) and WIHS (sample of target population)

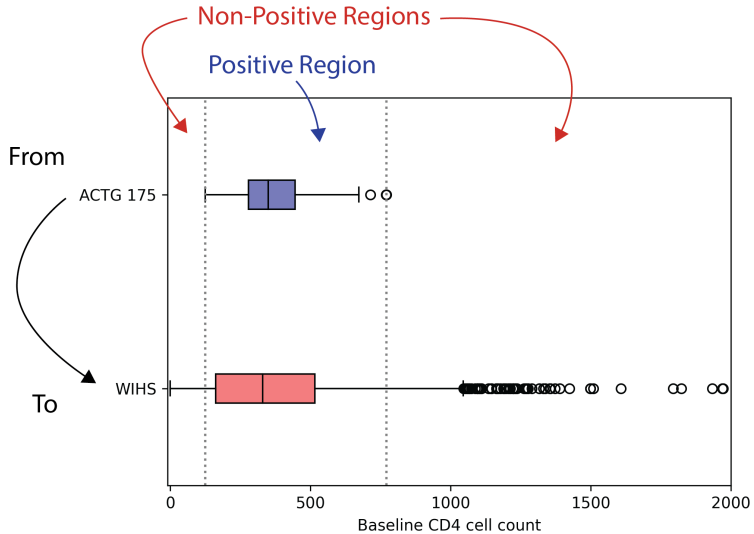
To transport assume populations are exchangeable given

- Baseline CD4²⁶

²⁵Case study from Zivich et al. *J R Stat Soc Series A* 2025; 188(1):158-180

²⁶This analysis also accounts for age, race, weight. Positivity was assumed for these variables

Distribution of Baseline CD4



Re-Expression of the Parameter

$$\psi = \underbrace{E[Y^1 - Y^0 \mid S = 1]}_{\text{Average causal effect (ACE) in population } S = 1} = E \left\{ \underbrace{E[Y^1 - Y^0 \mid V, S = 1]}_{\text{Conditional average causal effect (CACE) by } V} \mid S = 1 \right\}$$

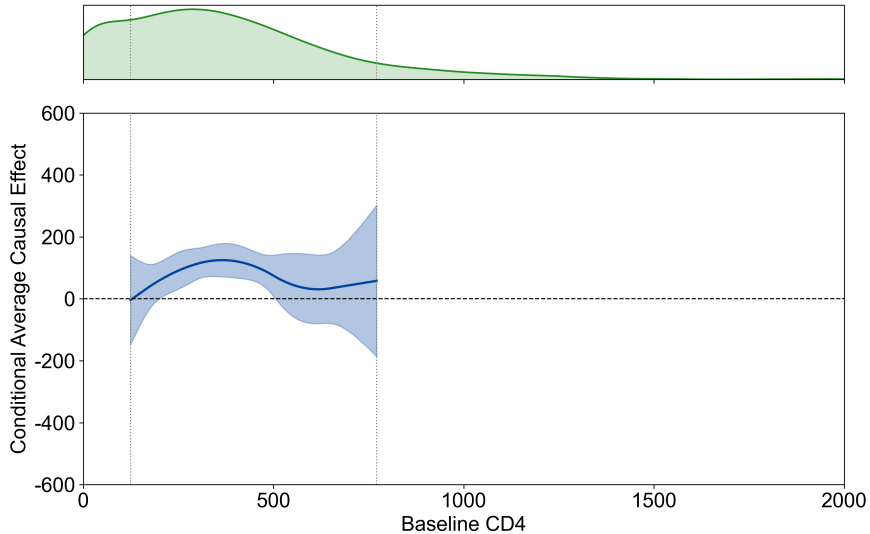
... or that the ACE is the CACE marginalized over the distribution of baseline CD4 values (i.e., V) in the target population (i.e., $S = 1$)²⁷

Therefore, we can look at the **CACE** for intuition on how non-positivity by V impacts estimation of the **ACE**²⁸

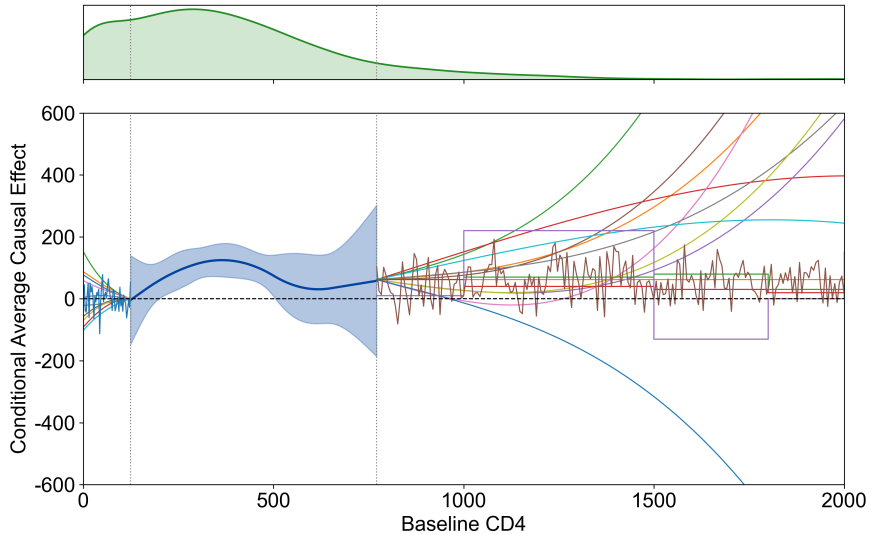
²⁷Where Y^a is the potential outcome under a and $E[\cdot]$ is the expected value function

²⁸For estimation of the CACE, I used an augmented inverse probability weighting estimator

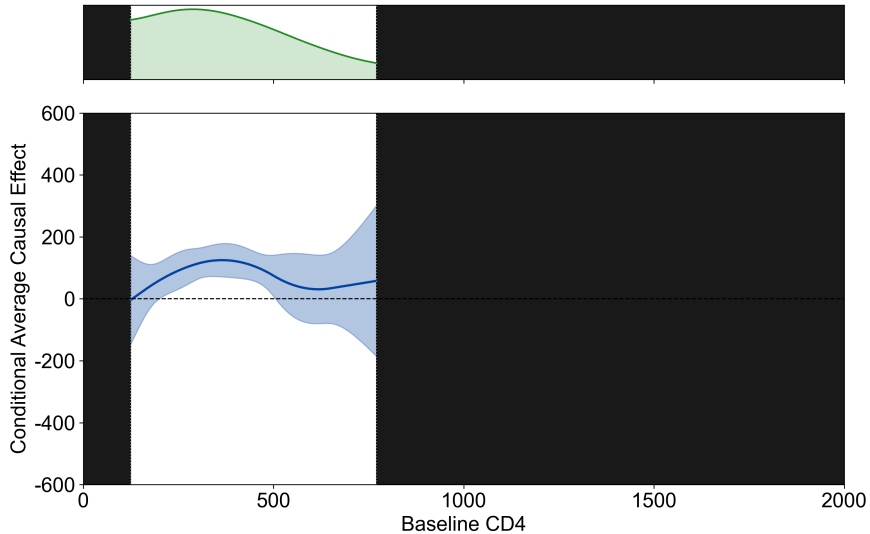
Conditional Average Causal Effect



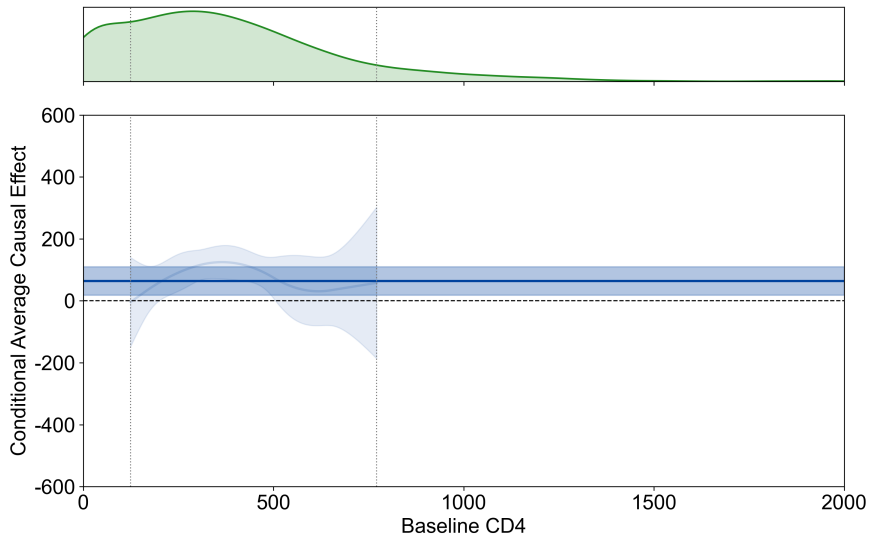
Challenge of Nonpositivity



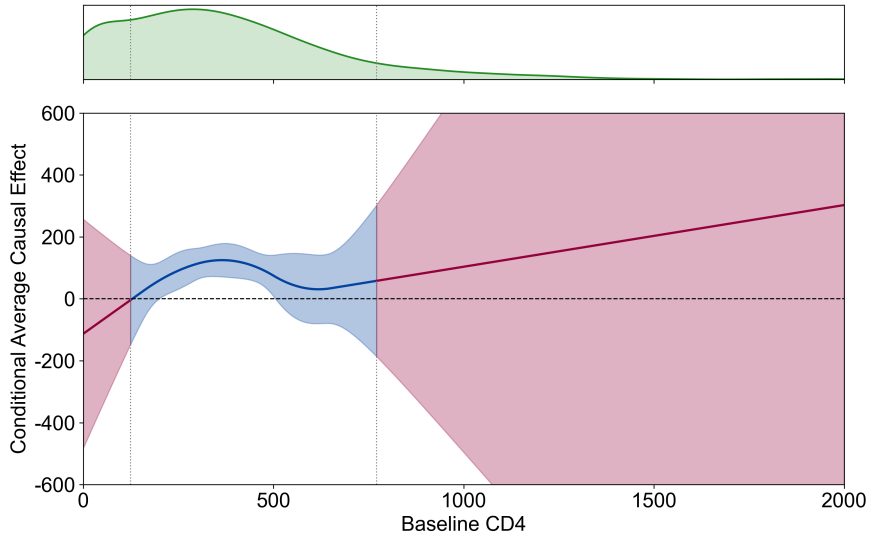
Solution 1: Restrict Target Population



Solution 2: Restrict Covariate Set



Solution 3: Extrapolate



Solution 4: Synthesis of Statistical and Mathematical Models

